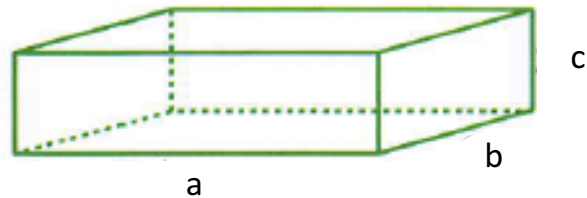


Year 11

Similarity worksheet - Introduction

1. What do you understand by similarity? Discuss the concept of similarity in Mathematics with your friend and jot down some ideas.

2. Consider the rectangular prism (cuboid) on the right. Using the information provided, calculate the surface area and volume of the prism for various dimensions shown in the table below.



Prism	1			2			3		
Dimensions of the prism (m)	a	b	c	a	b	c	a	b	c
	8	3	2	16	6	4	24	9	6
Surface area (m^2)									
Volume (m^3)									

3. Let us find the ratio of the dimensions, surface areas and volumes between Prisms 1 and 2 and between Prisms 1 and 3 and fill in the table below.

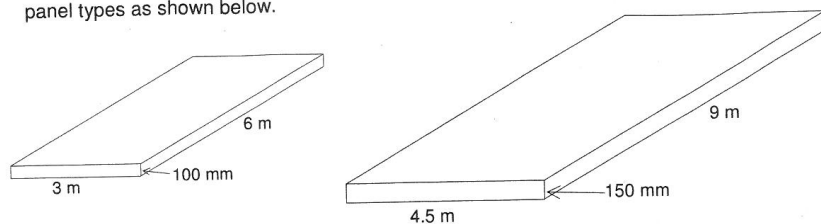
	Prisms 1 and 2	Prisms 1 and 3
Ratio of dimensions	1 : 2	1 : 3
Ratio of surface area		
Ratio of volumes		

4. What are your observations?

EXAMPLES

Example 1

A manufacturer of precast concrete panels used in the building industry manufactures two similar panel types as shown below.



- (a) The cost of concrete to make the smaller panel is \$280. Calculate the cost of concrete to make the larger panel

Before being used the panels require one coat of a special sealant which costs \$35/L. One litre of this sealant covers an area of 3.5 square metres.

- (b) Calculate the cost of sealing the smaller panel.
- (c) How much will it cost to seal the larger panel?

Example 2

When Emma was on holiday in Paris she purchase a replica of The Thinker, after visiting the Rodin Museum in Paris.

Her bronze replica was cast in the ratio 1:6. Emma's replica was 31 cm high and weighed 5 kg.

- (a) Calculate the height of the sculpture.

- (b) Estimate the weight of the sculpture.

- (c) What is the ratio of the surface area of her replica to that of the sculpture?

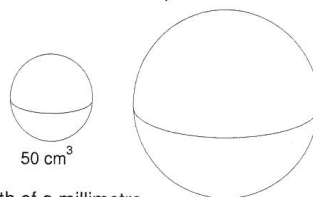


Example 3

The ratio of surface areas of two spheres is 1 : 16, and the volume of the smaller sphere is 50 cm^3 .

- (a) What is the ratio of the diameters of these spheres?

- (b) Calculate the volume of the larger sphere.



- (c) Calculate the radius of the smaller sphere to the nearest tenth of a millimetre.

- (d) Calculate the surface area of the larger sphere to the nearest square centimetre.